

Fit an sjSDM Model

Description

Fits a joint Species Distribution Model (jSDM) using the sjSDM package with specified abiotic, spatial, and error family configurations.

Usage

```
fit_jsdm_model(  
  data_to_fit = NULL,  
  sel_abiotic_formula = NULL,  
  abiotic_method = c("linear", "DNN"),  
  sel_spatial_formula = as.formula(~0 + coord_long:coord_lat),  
  spatial_method = c("linear", "DNN", "none"),  
  error_family = c("gaussian", "binomial"),  
  device = c("cpu", "gpu"),  
  parallel = 0L,  
  compute_se = FALSE,  
  biotic = NULL,  
  alpha_cov = 0.5,  
  alpha_coef = 0.5,  
  alpha_spatial = 0.5,  
  lambda_cov = 0,  
  lambda_coef = 0,  
  lambda_spatial = 0,  
  ...,  
  iter = 100L,  
  n_early_stopping = NULL,  
  verbose = FALSE  
)
```

Arguments

<code>data_to_fit</code>	A list containing the data to fit the model. Must include: - ' <code>data_community_to_fit</code> ': matrix of community composition (constant-presence taxa already removed by ' <code>filter_constant_taxa()</code> ') - ' <code>data_abiotic_to_fit</code> ': data frame of abiotic variables, already scaled by ' <code>scale_abiotic_for_fit()</code> ' - ' <code>data_spatial_to_fit</code> ': data frame of spatial predictors, already scaled by ' <code>scale_spatial_for_fit()</code> '. Required only when ' <code>spatial_method</code> ' is not <code>"none"</code> .
<code>sel_abiotic_formula</code>	A formula object specifying the abiotic (environmental) predictors
<code>abiotic_method</code>	Method for modeling abiotic effects. One of <code>"linear"</code> (default) or <code>"DNN"</code> (deep neural network)
<code>sel_spatial_formula</code>	A formula object specifying the spatial predictors. Defaults to an interaction between longitude and latitude (<code>~ 0 + coord_long:coord_lat</code>). Only used if ' <code>spatial_method</code> ' is <code>"linear"</code> or <code>"DNN"</code> .
<code>spatial_method</code>	Method for modeling spatial effects. One of <code>"linear"</code> (default), <code>"DNN"</code> (deep neural network), or <code>"none"</code> (no spatial structure)
<code>error_family</code>	Error family distribution. One of <code>"gaussian"</code> (default) or <code>"binomial"</code> . If <code>"binomial"</code> , community data is converted to presence/absence and a probit link is used
<code>device</code>	Computing device to use. One of <code>"cpu"</code> (default) or <code>"gpu"</code>
<code>parallel</code>	Number of CPU cores to use for parallel processing. Only applicable if ' <code>device = "cpu"</code> '. Default is <code>'0L'</code> (no parallelisation).
<code>compute_se</code>	Logical indicating whether to compute standard errors inline during model fitting. Default is <code>'FALSE'</code> . Prefer the post-hoc approach via ' <code>compute_jsdm_se()</code> ' in a separate pipeline target, which allows CPU parallelisation independent of the GPU device setting.
<code>biotic</code>	Optional biotic structure passed to ' <code>sjSDM::sjSDM()</code> '. Defaults to <code>'NULL'</code> , which creates ' <code>sjSDM::bioticStruct()</code> ' from the supplied ' <code>alpha_cov</code> ' and ' <code>lambda_cov</code> '. Use ' <code>sjSDM::bioticStruct(diag = TRUE)</code> ' to fit a no-associations variant.
<code>alpha_cov, alpha_coef, alpha_spatial</code>	Elastic-net mixing parameters for biotic association covariance, abiotic coefficients, and spatial coefficients. Each must be between zero and one, inclusive.

<code>lambda_cov,</code> <code>lambda_coef,</code> <code>lambda_spatial</code>	Non-negative regularization strengths for biotic association covariance, abiotic coefficients, and spatial coefficients.
<code>...</code>	Additional arguments passed to <code>'sjSDM::sjSDM()'</code> (e.g. <code>'sampling'</code> , <code>'step_size'</code> , <code>'seed'</code>). Do NOT pass <code>'iter'</code> , <code>'control'</code> or <code>'early_stopping'</code> directly — use the dedicated <code>'iter'</code> and <code>'n_early_stopping'</code> parameters instead.
<code>iter</code>	Positive integer. Number of training epochs. Default is <code>'100L'</code> .
<code>n_early_stopping</code>	Early stopping patience. Controls how many consecutive epochs without improvement are tolerated before training is halted. Three accepted values: - <code>'NULL'</code> (default): auto-compute as <code>'round(iter * 0.20)'</code> , i.e. at least 20 % of the epoch budget. - <code>'0L'</code> or negative integer: disables early stopping entirely. - Positive integer: uses <code>'max(value, round(iter * 0.20))'</code> to ensure patience is never set below 20 % of <code>'iter'</code> . Passed as <code>'early_stopping_training'</code> in <code>'sjSDM::sjSDMControl()'</code> .

Details

This function prepares the data and fits a joint Species Distribution Model using the `sjSDM` package. The spatial structure is modeled using an interaction term between longitude and latitude coordinates. When binomial error family is specified, the community data is converted to binary presence/absence data.

Standard error computation (`'compute_se = TRUE'`) may fail with certain model configurations, particularly when using DNN methods or complex spatial structures. If SE computation fails, the model will still be returned with a warning.

Value

An object of class `sjSDM` containing the fitted model

See Also

`sjSDM::sjSDM`, `sjSDM::linear`, `sjSDM::DNN`, `compute_jsdm_se`